Muhammad Irfan Fahmi

Computer Engineering | AI / ML Engineering
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Professional Summary

Computer Engineering background at UTeM with applied project work in object detection, hand-gesture classification and multi-model orchestration. Connects model inputs and evaluation to practical interfaces, data storage and observable system behaviour.

Education

Universiti Teknikal Malaysia Melaka (UTeM), Bachelor of Computer Engineering with Honours — Oct 2022 – Nov 2026

- Graduation scheduled: 2 November 2026.

Politeknik Port Dickson, Diploma in Electronic Engineering (Computer) — Dec 2018 – May 2022

Kolej Komuniti Selandar, Certificate in Computer Systems and Networking — Jul 2017 – Feb 2019

Technical Skills

Applied ML: YOLOv8, OpenCV, MediaPipe Hands, Random Forest, feature engineering, precision/recall, cross-validation.

Programming & integration: Python, JavaScript, SQL/MySQL, camera and barcode inputs, browser inference.

Agent systems: Multi-model consensus, LLM orchestration, Termux/Linux, telemetry, watchdog recovery.

Selected Projects

Hybrid Self-Checkout

- Built a Python pipeline combining YOLOv8/OpenCV product detection, USB barcode input and MySQL product/transaction data.
- Repository-reported evaluation: 77.4% precision and 72.0% recall after 50 epochs; tested visual verification challenges including occlusion and similar packaging.

Gesture Recognition / Touchless Reader

- Converted MediaPipe hand landmarks into 71-dimensional features and a 100-tree Random Forest for browser inference in JavaScript.
- Implemented gesture navigation with cooldowns and virtual controls. Reported 98.6% cross-validation accuracy; not a real-world user accuracy measurement.

Hermes / Multi-Model Agent Orchestration

- Built a documented Python/Termux experiment with multi-model consensus routing, telemetry, watchdog recovery and post-mortem analysis on mobile hardware.
- Separated serving and gateway processes and used telemetry to review model disagreement and unavailable-provider behaviour.

Technical Experience

Technical Staff, Global Elite Ventures Sdn. Bhd — Sep 2021 – Jul 2022

Technical troubleshooting and hardware/software deployment.

Contract Assistant Engineer, ARNN Technologies & Karar Solution — Jun 2020 – Jul 2020

Engineering project support and technical problem-solving.

Technician, OKCS Seri Kembangan — Jun 2018 – Nov 2018

Diagnostics, repairs, hardware maintenance and software configuration.

Professional Registrations

BEM: Graduate Engineer, Computer branch; approved 15 October 2026. Public directory.

MBOT: Graduate Technologist, Information & Communication Technology (IT); GT26100872; approved 22 October 2026.

Selected Training

Festo / STDC: Industrial Automation with AI in Manufacturing, 14-16 July 2026.

Cisco Networking Academy: Enterprise Networking, Security, and Automation; Switching, Routing, and Wireless Essentials (23 February 2026). Course completions.

Languages

Bahasa Malaysia (native); English (professional working proficiency).